



S. A. ENGINEERING COLLEGE, CHENNAI-77

(Autonomous-Institute Level Research Centre-Affiliated to Anna University)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution
Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.

ACTION TAKEN FOR OVERALL IMPROVEMENTS AFTER THE PREVIOUS NBA VISIT

Strengths observed by committee:

1. Vision and Mission statements are well defined and consistent with institute vision and mission
2. Enrolment ratio of students is good
3. Placements are of good quantity
4. Faculty cadre proportion, faculty participation in FDPs are good
5. Faculty retention is good

Weakness/ Areas of Improvement observed by committee

Sl No	Previous NBA visit comments	Action taken for Improvements
1.	Quality of Internal question papers needs improvement	➤ A Question Paper Blueprint is developed for each course before every internal assesment test defining the provision of COs, Bloom's Levels, and mark distribution. ➤ The question paper includes part-A short answers and part-B/C problem-solving or case study questions. ➤ The question papers are moderated and verified against the blueprint by the Scrutiny Committee consisting of External and Internal Professors/Senior faculty and approved by HoD before submitting to Exam Cell.

2.	OBE understanding needs improvement among stakeholders	All faculty members have completed the NPTEL certificate on ➤ Accreditation and Outcome based Learning ➤ NBA Accreditation and Teaching-Learning in Engineering (NATE) Faculty Development training programs organized/attended by the department by the faculty ➤ 1 week FDP on “National Education Policy and Outcome Based Education” from 26/12/2024 to 31/12/2024 Sponsored by IEEE Communication Society Madras Chapter ➤ 1 day workshop attended by Dr.R.Geetha on “NBA Awareness Workshop on Outcome Based Education and Accreditation” organized by Anna University on 09-02-2025 (Dr. Anil Kumar Nassa, Member Secretary, Prof. Anil D. Sahasrabudhe, Chairman, NBA, Prof. S.S. Pattnaik, Vice Chancellor, Odisha State Open University, Prof. R.V. Ranganath, Dean (Academics), M.S. Ramaiah University of Applied Sciences, Bangalore) ➤ 2 days FDP on “Outcome Based Education (OBE) New Aspects of NBA Accreditation and Assessment Practices” on 30-08-2025 & 06-09-2025 by Dr. S. Muthan, Professor of Eminence, Anna University Chennai. ➤ 1 day southern zone conference for Autonomous colleges on “Implementation of NEP 2020” 14-11-2024 by IITM & UGC ➤ 1 day workshop on “Implementation of OBE “ 20-07-2024 by Dr. Gnanasambanthan, Professor-NITTR ➤ 1 day national Seminar on “ National Education Policy 2020” - 12-02-2024 by Dr.Prasanth Nair -IQAC Dean - Amrita Vishwa vidyapeetham ➤ 1 day workshop on “Outcome Based Education” On 21-10-2023 by Dr.Thiyagarajan & Dr.Muttan, Anna university ➤ 1 day workshop on “Co-PO Mapping and Attainment of POs” On 28-10-2023 by Dr.Thiyagarajan & Dr.Muttan, Anna university Awareness on OBE to Students ➤ Every Semester before the start of syllabus the OBE and CO-PO mapping are explained in class by the concerned subject in charge ➤ Students' knowledge on OBE is verified through quiz.
3.	Quality of Placements need improvement	Training to Improve the Quality of Placements To enhance the overall placement quality, various structured training programs are implemented: ➤ Communication Skills: A Mentor–Mentee Model is followed using PEARSON MEPRO to strengthen students' communication and language proficiency. ➤ Aptitude Skills: To develop analytical and logical reasoning, aptitude training and practice sessions are conducted under the Mentor–Mentee Model using Aptitude Buster. ➤ Technical Skills: Technical competency is improved through the Mentor–Mentee Model facilitated by SkillRack. ➤ Motivational and Certification Support: Students are encouraged and guided to pursue industry-recognized certifications such as AWS

		Certification, Infosys Springboard, and Wipro Certified Faculty Training Programs, which are also extended to faculty members to enhance mentoring quality. Industry-Based Training and Placement Initiatives To enhance students' employability and exposure to real-time industry practices, industry-based training programs are organized regularly. In addition, reputed and high-package companies such as AVASOFT, TCS, WIPRO, and KAAR Technologies are invited to conduct campus recruitment drives, providing students with excellent career opportunities in leading industries.
4.	Computing servers needs to provided more	1. One Lenovo ThinkSystem SR650 V3 2U rack server is equipped with a single Intel Xeon Gold 5418Y processor (24 cores, 2.0GHz, 45MB cache, 185W TDP). It has 64GB of DDR5 RAM (2 x 32GB) and five 960GB SATA SSDs installed in an 8-bay 2.5" SATA/SAS backplane. The server includes a RAID 9350-8i controller with 2GB flash cache and is managed through the XCC Platinum system. Networking is provided by an OCP Card 57416 with dual 10GbE 10GBASE-T ports. Power is supplied by two 750W Platinum-rated (230V/115V) Gen 2 v3 power supplies. This server is configured as a file server for student use. 2. Two Intel Core i9 14th generation desktop servers are provided for the Operating System and Networking Lab, configured with a Linux server to enable Telnet access. Additionally, a dedicated database server is available to access Oracle 19c. 3. One Lenovo ThinkSystem SR650 (Type 7X06, Serial Number J300CHWT) serves as a file server for faculty members.
5.	Sponsored research, academic research needs improvement	Three Sponsored Research Grants of worth Rs 89,95,764 received from DST and MeiT - Quantum machine learning algorithms for traffic management in big cities from MeitY Quantum Computing Applications Lab (QCAL) worth of Rs 7,95,810 (USD: 9705) - An ontology-based 3D digital documentation of Masilamaniswara Temple integrating techniques of terrestrial laser scanning and unmanned aerial vehicle digital photogrammetry from Science and Heritage Research Initiative (SHRI) Department of Science and Technology (DST) worth of Rs.33,70,020 - Science Technology and Innovation Hub for enhancing employability & livelihood of SC Community Population through technology development of coir-based products for application in construction & Manufacturing from STI Hub worth of Rs 48,29,934.

		Academic Research From 2022 to till date, a total of 198 research publications among that 50 papers in SCI-indexed journals and 148 in Scopus-indexed journals also filed 55 patents, with 2 patents granted, reflecting a strong emphasis on innovation and applied research. Among these, 9 papers were published in Q1 category journals and 24 in Q2 category journals. In the academic year 2025–26, 27 scholars are enrolled in Ph.D. programs.
6.	Success rate without backlogs needs improvement	Percentage of improvement is 9% Previous Success rate without backlog - 38% (9.6 out of 25) Current Success rate without backlog - 47% (7.05 out of 15)

Deficiency

Sl No	Previous NBA visit comments	Action taken for Improvements
1.	No consultancy during assessment period	For the Current academic year 2025–26, two ongoing consultancy projects are being carried out in collaboration with Webzenith Solutions, each valued at Rs. 1,00,000. ➤ The first project, titled “AI-Powered Contextual Learning Platform,” is led by Dr. R. Geetha and Dr. N. Thilagavathi, ➤ The second project, titled “Real-Time Interactive Live-Streaming Platform,” is undertaken by Dr. M. Anuradha and Dr. V. Kaliraj. 2024–25 1. Advanced AI Training was conducted in collaboration with IIT Delhi’s Sonipat by Dr. R. Geetha, with a consultancy value of Rs 30,000 for one day (18.11.2024). 2023–24 1. Interactive Learner’s Portal was developed in collaboration with Techregech Cyberlabs Pvt. Ltd., coordinated by Dr. R. Geetha and Dr. D. Chitra Devi, with a consultancy worth Rs 2,00,000 for six months. 2. Building LMS Portal was executed with Beeja Academy, coordinated by Dr. M. Anuradha and Dr. D. Hemanand, valued at Rs 2,00,000 for six months. 3. Data-based Modeling using Machine Learning was carried out in partnership with Altimetrik, coordinated by Dr. M. Nillani and Dr. A. Anto Udoss, with a project cost of Rs 1,00,000 for six months. 4. Volumetric Segmentation using Mosaic was done with Satchak, coordinated by Mr. M. Balasubramanian, costing

Rs 87,500 for six months.

5. React JS Website for Microfab was completed in collaboration with **Anacle Studios**, coordinated by **Dr. M. Thilagavathy**, valued at **Rs 80,000 for six months.**
6. Bond Management System using Mendix was executed with **Rapid Data Technologies**, coordinated by **Mr. C. Balakrishnan**, with a consultancy cost of **Rs 50,000 for six months.**
7. Merchant Management System was done in collaboration with **Dave AI**, coordinated by **Dr. S. Sivakumar** and **Mr. S. Muthukumarasamy**, valued at **Rs 32,448 for six months.**

2021-22

1. Correcting Improperly White-Balanced Image was carried out in collaboration with **GOX-AI Company**, coordinated by **Dr. R. Geetha** and **Mr. C. Balakrishnan**, with a consultancy value of **Rs 3,000 for a duration of three months.**
2. Portrait Shadow Manipulation was executed with **GOX-AI Company**, coordinated by **Dr. R. Geetha** and **Mr. C. Balakrishnan**, valued at **Rs 3,000 for a duration of three months.**

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